(b) Non-parametric form of vector equation

Let $\vec{r}=\vec{a}+s\vec{b}$ and $\vec{r}=\vec{c}+t\vec{d}$ be two non-parallel coplanar lines. Then $\vec{b}\times\vec{d}\neq\vec{0}$. Let P be any point on the plane and let $\vec{r_0}$ be its position vector. Then, the vectors $\vec{r_0}-\vec{a},\vec{b},\vec{d}$ as well as $\vec{r_0}-\vec{c},\vec{b},\vec{d}$ are also coplanar. So, we get $\left(\vec{r_0}-\vec{a}\right).\left(\vec{b}\times\vec{d}\right)=0$ or $\left(\vec{r_0}-\vec{c}\right).\left(\vec{b}\times\vec{d}\right)=0$. Hence, the vector equation in non-parametric form is $\left(\vec{r}-\vec{a}\right).\left(\vec{b}\times\vec{d}\right)=0$ or $\left(\vec{r}-\vec{c}\right).\left(\vec{b}\times\vec{d}\right)=0$.

(C) Cartesian form of equation of plane

In Cartesian form the equation of the plane containing the two given coplanar lines

$\frac{x-x_1}{b_1}=\frac{y-y_1}{b_2}=\frac{z-z_1}{b_3}$ and $\frac{x-x_2}{d_1}=\frac{y-y_2}{d_2}=\frac{z-z_2}{d_3}$ is given by

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ b_1 & b_2 & b_3 \\ d_1 & d_2 & d_3 \end{vmatrix} = 0 \quad \text{or}$$

$$\begin{vmatrix} x-x_2 & y-y_2 & z-z_2 \\ b_1 & b_2 & b_3 \\ d_1 & d_2 & d_3 \end{vmatrix} = 0$$

Example 6.46

Show that the lines $\vec{r}=\left(-\hat{i}-3\hat{j}-5\hat{k}\right)+s\left(3\hat{i}+5\hat{j}+7\hat{k}\right)$ and $\vec{r}=\left(2\hat{i}+4\hat{j}+6\hat{k}\right)+t\left(\hat{i}+4\hat{j}+7\hat{k}\right)$ are coplanar. Also,find the non-parametric form of vector equation of the plane containing these lines.

Solution

Comparing the two given lines with

$$\vec{r}=\vec{a}+t\vec{b},\ \vec{r}=\vec{c}+s\vec{d}$$

we have, $\vec{a}=-\hat{i}-3\hat{j}-5\hat{k},\ \vec{b}=3\hat{i}+5\hat{j}+7\hat{k},\ \vec{c}=2\hat{i}+4\hat{j}+6\hat{k}$ and $\vec{d}=\hat{i}+4\hat{j}+7\hat{k}$

We know that the two given lines are coplanar , if $\left(\vec{c}-\vec{a}\right)\cdot\left(\vec{b}\times\vec{d}\right)=0$

Here, $\vec{b}\times\vec{d}=\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 5 & 7 \\ 1 & 4 & 7 \end{vmatrix}=7\hat{i}-14\hat{j}+7\hat{k}$ and $\vec{c}-\vec{a}=3\hat{i}+7\hat{j}+11\hat{k}$

Then, $\left(\vec{c}-\vec{a}\right)\cdot\left(\vec{b}\times\vec{d}\right)=\left(3\hat{i}+7\hat{j}+11\hat{k}\right)\cdot\left(7\hat{i}-14\hat{j}+7\hat{k}\right)=0$.

Therefore the two given lines are coplanar.Then we find the non parametric form of vector equation of the plane containing the two given coplanar lines. We know that the plane containing the two given coplanar lines is

$$\left(\vec{r}-\vec{a}\right)\cdot\left(\vec{b}\times\vec{d}\right)=0$$

which implies that $\left(\vec{r}-\left(-\hat{i}-3\hat{j}-5\hat{k}\right)\right)\cdot\left(7\hat{i}-14\hat{j}+7\hat{k}\right)=0$. Thus, the required non-parametric vector equation of the plane containing the two given coplanar lines is $\vec{r}\cdot\left(\hat{i}-2\hat{j}+\hat{k}\right)=0$.

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